IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2022

COMMUNICATION NETWORKS

Friday, 16 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : J.A. Barria
Second Marker(s) : C. Ling

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Information for students

1. Mean delay for the M/M/1 system may be taken as

$$t_i = \frac{1}{\mu C_i - \lambda_i}$$

where,
$1/\mu$ = Average length of packet [bit/packet]
C_i = Transmission speed link i [bits/s]
μC_i = Service rate (link i) [packet/ s]
λ_i = Arrival rate (link i) [packet/ s]

2. Optimal Routing Problem (ORP)

$Min\ D(F)$ with respect to $F = \{F_i\}$

where, $D(F) = \sum_{i=1}^{L} \frac{F_i}{C_i - F_i}$

and,

C_i = Capacity of link l_i.
F_i = Flow carried by link l_i.

3. Velocity of propagation and propagation delay.

Velocity of unguided transmission through air or space of an electromagnetic wave is $\mathbf{2.99x10^8}$ m/s

Assume velocity of guided transmission to be approximately $\mathbf{66\%}$ of the unguided wave propagation.

And propagation delay:

$$\boldsymbol{a = \frac{Rd}{LV}}$$

Where,
$\boldsymbol{R}$ = Data rate of the channel.
$\boldsymbol{d}$ = Length of the network.
$\boldsymbol{L}$ = Average frame length.
$\boldsymbol{V}$ = Velocity of guided media.

The Questions

Q1.

a) The sequence in **Fig. 1.1.** shows the effect of the propagation (a) on the utilization of a ring network. Note: 1 is the normalised transmission time and a is the propagation delay (normalised):

i) at which time stamp the free token could be sent, and at what time is actually sent (for $a \geq 1$ and $a < 1$) respectively. **[4]**

ii) using the set of values of a representative of a ring in **Table. 1.1.**: calculate the utilization of the network and discuss your fundings. **[4]**

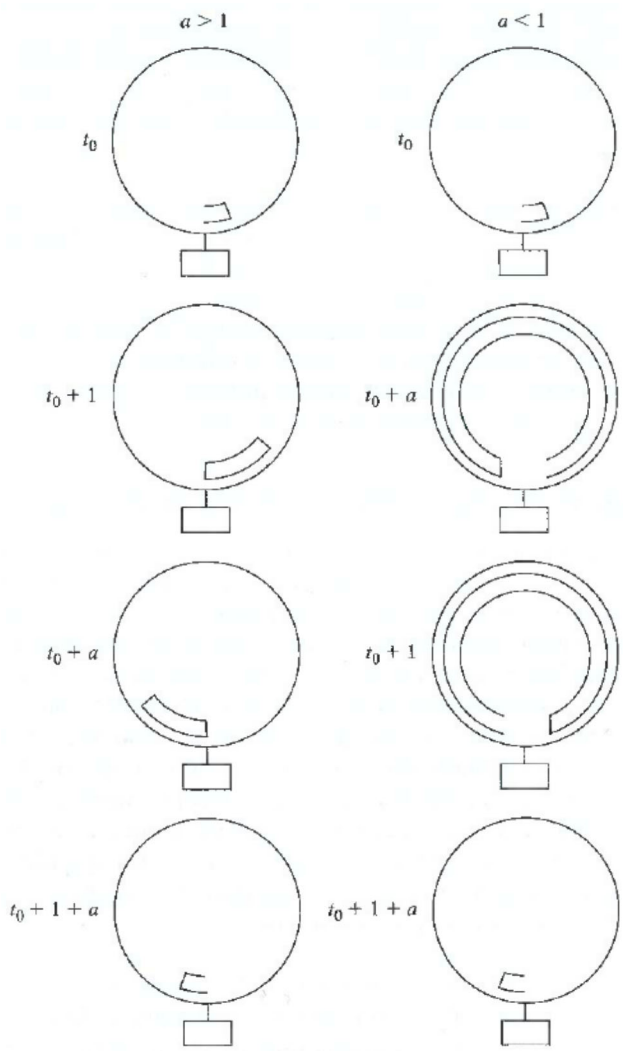


Figure 1.1.: Ring Network (**Ref.**: Stallings W. "Data and Computer Communications", NY. MacMillan)

Table 1.1: Network characteristic features

Data Rate	Frame size	Network length
Mbps	Bits	Km
1	100	1
1	100	10
10	1000	10
100	1000	50

b) For the go back N ARQ mechanism, if a frame is in error the average numbers of retransmissions N_r can be expressed as:

$$N_r = \frac{1 - P + KP}{1 - P}$$

where, P is the probability that a single frame is in error and K is an estimation of the number of frames that will require retransmission if one frame is in error.

i) Estimate the value of K when the window size $W \geq 2a + 1$ and when $W < 2a + 1$ **[4]**

Hint: you can make use of sequence diagrams if necessary.

c) Internet mobility

i) Briefly state all the elements that an Internet host requires to connect to the Internet. **[3]**

ii) In the context of Internet mobility explain the usefulness of the DHCP protocol. **[5]**

< END OF QUESTION 1 >

Q.2.1.

i) Introduce and discuss the difference between a traditional shortest path problem and a statistical shortest path problem. Use examples to highlight the differences. **[4]**

ii) Discuss the underlying purpose of the Internet ICMP protocol and briefly explain how the protocol information is communicated between routers. **[4]**

Q.2.2.

The total energy consumption of a K node wireless network is given by the following expression:

$$\text{Total Power} = \sum_{k=1}^{k} (E_t + E_r)\, B$$

For the transmitter: $E_t = (e_t + e_d r^2)B$.
For the receiver: $E_r = e_r B$.

where,
$e_t = e_r = 5 \cdot 10^{-10}$ Jules/bit.
$e_d = 100 \cdot 10^{-12}$ Jules/bit/m^2.
$B = 2$ MBit/s.
$d =$ distance in m between communicating nodes.

i) At what distance the e_d energy consumption becomes dominant contributor to the Total Power consumption. **[3]**

ii) Assuming that e_t and e_r are negligible in comparison to e_d, and using the distances shown in **Fig. 2.1.**: discuss the attributes of the alternative paths and the one you would select to transmit. **[5]**

iii) If you are given the option to move (or place) a new node in **Fig. 2.1.**, where will you place this node? Clearly argue your solution. **[4]**

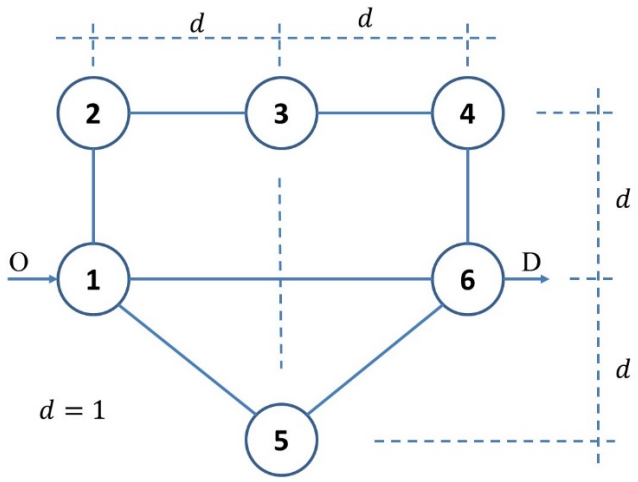


Figure 2.1: Wireless network.

< END OF QUESTION 2 >

Q.3.1. Provision of multiple classes of services

i) Briefly introduce and discuss four (4) network mechanisms known to you to support multiple classes of services on a packet network. **[8]**

Q.3.2.

Consider the network in **Fig. 3.1.** with the following link costs $l(i)$:

$$l(1) = \frac{x(1)}{C(1) - x(1)}$$

$$l(2) = \frac{x(2)}{C(2) - x(2)}$$

$$l(3) = \frac{a}{x(1) + x(2)}$$

where a, is a system parameter and $x(1) and\ x(2)$ are the corresponding traffic flows being carried by link $l(1) and\ l(2)$ respectively.

Note: $C(1) > C(2)$ and $C(1) + C(2) \geq x(1) + x(2)$.

i) Derive a formal expression for the network cost function as an additive separable function of the link costs. **[2]**

ii) Discuss in your own words what is the effect of including the link cost $l(3)$. Write down the condition for link $l(1), l(2)\ and\ l(3)$ to carry traffic flows. **[5]**

iii) Considering that $C(1) > C(2)$: calculate the magnitude of the flow $x(1)$ that link $l(1)$ can carry before the other links (link $l(2)$ and link $l(3)$) start carrying flow. **[5]**

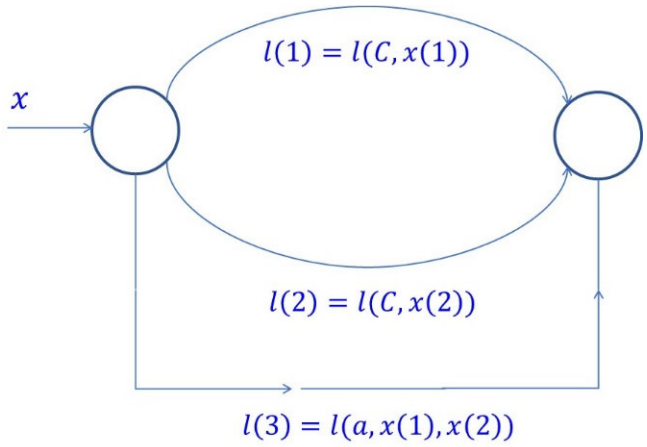


Figure 3.1: Network of Q3.2.

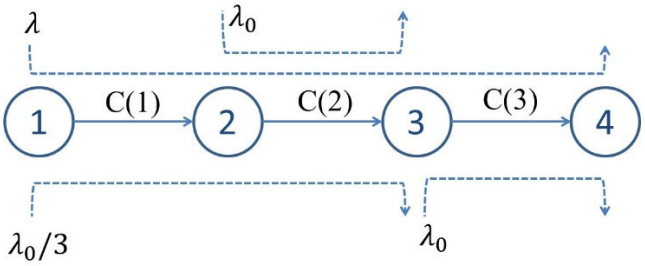
< END OF QUESTION 3 >

Q.4.1.

In the context of a network utility maximisation (NUM) problem:

i) Introduce and explain the idea behind the proportional fair allocation principle. **[3]**

ii) Using the network sown in **Fig. 4.1.** solve the flow assignment problem using proportional fair allocation. **Hint:** Identify the bottle neck link before assigning the flows. **[6]**



C(1) = C(2) = C(3) = C = 1 (link Capacity)

Figure 4.1: Proportional fair flow allocation network.

Q.4.2.

a) In the context of the 802.5 (WiFi) protocol:

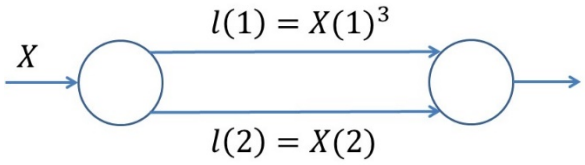
i) Briefly discuss the rational behind proposing a RTS-CTS sequence. **Hint:** Use time diagram if necessary. **[2]**

ii) Discuss the usefulness of the RTS-CTS sequence in respect to the underlying traffic load conditions and in respect to the average frame size to be transmitted. **[3]**

b) In the context of an ORP. For the network presented in **Fig. 4.2.**:

i) Given the link costs $l(1)$ and $l(2)$: derive the optimality condition. **[2]**

ii) Sketch a ***flow_carried*** vs ***flow_offered*** graph clearly identify in the x axis both $x(1)$ and $x(2)$ flows. **[4]**



$X = X(1) + X(2)$

Figure 4.2.: ORP network.

< END OF QUESTION 4 >